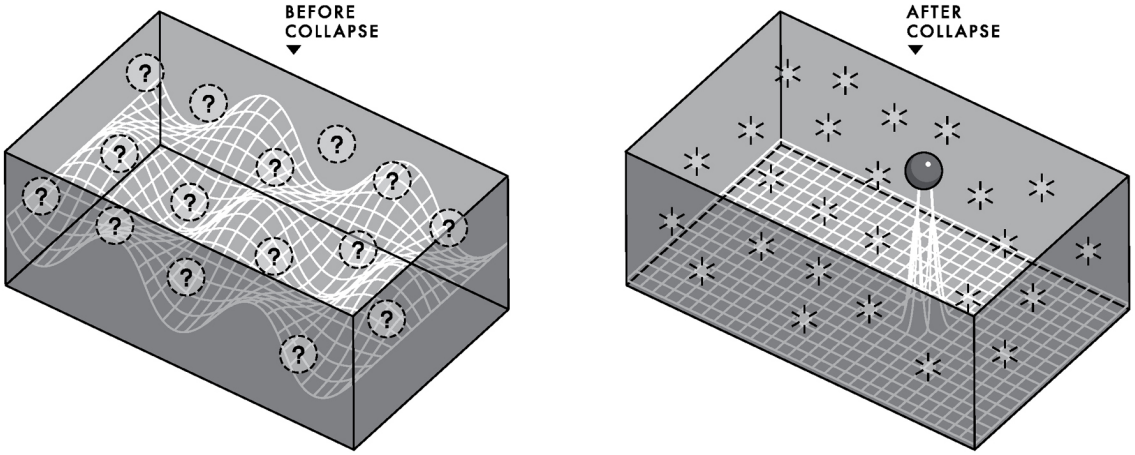


Figure 1.1. The measurement problem. Left: The wave function of a ball in a box undulates smoothly, like ripples on the surface of a pond, governed by the Schrödinger equation. The ball could be anywhere in the box. Right: The ball's location is measured and found in a particular spot. The wave function immediately and violently collapses, radically disobeying the Schrödinger equation. Why does the Schrödinger equation—a law of nature—apply only when measurements are not occurring? And what counts as a "measurement" anyhow?

Why does the Schrödinger equation only apply when measurements aren't happening? That doesn't seem to be how laws of nature work—we think of laws of nature as applying all the time, no matter what we're doing. If a leaf detaches from a maple tree, it will fall whether or not anyone is there to see it happen. Gravity doesn't care whether anyone is around to watch.

But maybe quantum physics really is different. Maybe measurements do change the laws that govern the quantum world. That's certainly strange, but it doesn't seem impossible. But even if that's true, it still doesn't solve the measurement problem, because now we have a new challenge: what is a "measurement," anyhow? Does a measurement require a measurer? Does the quantum world depend on whether it has an audience? Can anyone at all collapse a wave function? Do you need to be awake and conscious for it, or can a comatose person do it? What about a newborn baby? Is it limited to humans, or can chimps do it too? "When a mouse observes, does that change the [quantum] state of the universe?" Einstein once asked. Bell